

# Homework 8

## STAT 300 (Fall 2024)

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**Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx\_Yourfirstname.nb.html> file by the deadline.**

(1). A *Clothing* data set in R package *Stat2Data* contains data from a random sample of 60 customers from a large clothing store. The store manager is interested in understanding how much a customer will spend on his or her new purchase based on available characteristics.

The data dictionary is as follows:

- *ID* - Customer ID
  - *Amount* - Net dollar amount spent by customers in their latest purchase from this retailer
  - *Recency* - Number of months since the last purchase
  - *Freq12* - Number of purchases in the last 12 months
  - *Dollar12* - Dollar amount of purchases in the last 12 months
  - *Freq24* - Number of purchases in the last 24 months
  - *Dollar24* - Dollar amount of purchases in the last 24 months
  - *Card* - whether or not the customer has a private-label credit card with the retailer (1=yes, 0=no)
- (a) Install the R package and access the Clothing data. Print the first 5 lines of your data.
- (b) The manager mentioned that if the customers purchased the items and then returned them, the net dollar amount spent by customers in their latest purchase will be marked as zero. He said he is not interested in returns, so we should remove such customers from the analysis. Identify the customers who have returned their purchases, and remove them from the data. How many customers are remaining in your data after removing these customers? You may use the *subset()* function and *dim()* function to answer this question. You will be moving forward with this reduced dataset.
- (c) Remove the ID column from the data as it is not a variable and simply an identifier for subjects. Show the first 5 observations after removing ID column.
- (d) Obtain the correlation matrix and the scatterplot matrix. Comment on your observations.
- (e) Are there any predictors that are correlated with other predictors? If so, what are those predictors that can potentially create a multicollinearity issue in a fitted model.
- (f) Use the best subset method with BIC to come up with a reasonable model. What are the predictors suggested to be included in the model.

- (g) Once the model in part (f) was presented to the manager, he suggested including *Freq12* instead of *Freq24* and *Dollar12* instead of *Dollar24* as they are highly correlated. Obtain the least squares regression line to explain net dollar amount spent by customers in their last purchase using *Dollar12*, *Freq12* and *Card* . Write down the estimated equation for this model.
  - (h) Obtain the model diagnostics for the 3 predictor model. Discuss the validity of the model assumptions and presence of any unusual points. If you have unusual points also see if any of them could be influential.
  - (i) A quick consultation with the manager reveled that the data record indicating a customer has spent \$1,506,000 in the store is a data entry error and therefore should be removed from the analysis. Remove this data point. How many records are in your dataset now?
  - (j) During this discussion, it was also decided that the average amount spent on each visit by the customer would be a better predictor to consider than total dollar mount of purchases. Therefore instead of using *Freq12* and *Dollar12* as predictors, it was decided to use  $AvgSpent12 = \frac{Dollar12}{Freq12}$ . Keep in mind that when constructing the new variable, you will need to pay attention to limitations in the data that could cause potential issues (for example, what if you divide by zero). After removing any records that may create such issues, how many records you have in the data now? Print the first 3 observations of the data with the newly created variable.
  - (k) Re-fit the model with predictors *AvgSpent12* and *Card*, and obtain the model diagnostics for the 2-predictor model. Discuss the validity of the model assumptions and presence of any unusual points.
  - (l) Take the natural log transformation of both the *AvgSpent12* and *Amount*. Fit a model of  $\log(\text{Amount})$  on  $\log(\text{AvgSpent12})$  and *Card*. Label this 2-predictor model as  $M_1$ . Discuss the validity of the model assumptions and presence of any unusual points for  $M_1$ .
  - (m) Write down the estimated equation for model  $M_1$ . Are there any predictors that we can remove at 5% level? If so, what are they? Explain your reasoning.
  - (n) Interpret the estimated coefficient for variable *Card* in the context of the problem.
  - (o) Model the log-transformed response using only the log transformed *AvgSpent12*. Label this model as  $M_2$ . Would you prefer model  $M_2$  over model  $M_1$ ? Explain your answer.
- (2). State if below statements are true or false.
- (a) Randomly selecting units from populations and performing an ANOVA allow you to generalize the findings to the populations where the data came from.
  - (b) In the datasets in which there is no random selection, we can generalize the findings from the ANOVA results to the populations.
  - (c) If the dataset does not meet the independence condition for the ANOVA model, a transformation can improve this.

- (3). The main purpose of a one-way ANOVA model is to learn about:
- (a) the variances of several groups.
  - (b) the means of several groups
  - (c) the mean of one group
  - (d) the variance of one group
- (4). Which of the following is not a condition of the ANOVA model
- (a) Error terms have the same standard deviation.
  - (b) Error terms are all positive
  - (c) Error terms are independent
  - (d) Error terms follow a normal distribution
- (5). The best two plots to check the validity of ANOVA assumptions are:
- (a) Bart chart of explanatory groups and normal probability plot.
  - (b) Normal probability plot of the residuals and scatterplot of residuals vs. fitted values.
  - (c) Normal probability plot of the residuals and dotplot of the residuals.
  - (d) Scatter plot of the response vs. predictor and dotplot of residuals.